STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define period. [1]
2. Frequency-period relation? [1]
3. Restoring force in SHM? [1]
4. Where is speed maximum in SHM? [1]
5. Where is acceleration magnitude maximum? [1]
6. Spring angular frequency? [1]
7. Small-angle pendulum period? [1]
8. Total spring energy for amplitude A? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. Period .5s: frequency? [2]
10. m=.5kg,k=200N/m: period? [2]
11. For x=Acosomegat, acceleration? [2]
12. Pendulum length becomes 4l: period factor? [2]
13. Velocity at an extreme? [2]
14. Condition for pendulum SHM approximation? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. m=.5,k=200: angular frequency and period? [3]
16. A=.1m,omega=4: maximum speed and acceleration? [3]
17. Pendulum length 1m,g=9.8: period? [3]
18. Describe energy in one SHM cycle. [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. k=50N/m,A=.2m: total energy and U at x=.1m? [4]
20. Why does a real oscillator stop? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- Frequency f=1/T; angular frequency omega=2pi f=2pi/T.
- SHM equation a=-omega^2 x; spring force F=-kx.
- For a spring-mass oscillator omega=sqrt(k/m), T=2pi sqrt(m/k).
- For a simple pendulum at small angles T=2pi sqrt(l/g).
- Maximum speed v_max=omega A; maximum acceleration a_max=omega^2 A.
- Total energy E=(1/2)kA^2; potential energy U=(1/2)kx^2; kinetic energy K=E-U.
- Period scaling: spring T proportional to sqrt(m/k); pendulum T proportional to sqrt(l/g).
- Undamped SHM exchanges kinetic and potential energy while total mechanical energy remains constant.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Time for one cycle.
Final answer: Time for one cycle.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: f=1/T.
Final answer: f=1/T.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: F=-kx.
Final answer: F=-kx.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: At equilibrium.
Final answer: At equilibrium.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: At extreme displacement.
Final answer: At extreme displacement.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: sqrt(k/m).
Final answer: sqrt(k/m).

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 2pisqrt(l/g).
Final answer: 2pisqrt(l/g).

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: kA^2.
Final answer: kA^2.

Question 9

Working: f=1/T=1/0.5=2 Hz.

Final answer: 2 Hz.

Question 10

Working: T=2pi sqrt(m/k)=2pi sqrt(0.5/200)=2pi sqrt(0.0025)=2pi(0.05)=0.314 s.

Final answer: 0.314 s.

Question 11

Working: For x=A cos(omega t), acceleration is the second derivative: a=-omega^2 A cos(omega t)=-omega^2 x.

Final answer: a=-omega^2x.

Question 12

Working: Pendulum period T=2pi sqrt(l/g). If length becomes 4l, T changes by sqrt(4)=2, so period doubles.

Final answer: 2.

Question 13

Working: At an extreme, displacement magnitude is A and all ideal SHM energy is potential, so instantaneous velocity is zero.

Final answer: Zero.

Question 14

Working: For the small-angle approximation, the angular displacement must be small and measured in radians; then sin(theta) is approximately theta and restoring torque is proportional to angle.

Final answer: Small angular displacement.

Question 15

Working: omega=sqrt(k/m)=sqrt(200/0.5)=sqrt400=20 rad/s. T=2pi/omega=2pi/20=0.314 s.

Final answer: omega=20 rad/s; Tapprox 0.314 s.

Question 16

Working: v_max=A omega=0.1(4)=0.4 m/s. a_max=A omega^2=0.1(16)=1.6 m/s^2.

Final answer: 0.4 m/s and 1.6 m/s^2.

Question 17

Working: T=2pi sqrt(l/g)=2pi sqrt(1/9.8)=2.01 s.

Final answer: approx 2.01 s.

Question 18

Working: At equilibrium, speed and kinetic energy are maximum while spring/gravitational potential energy is minimum. At extremes, speed is zero and potential energy is maximum. In undamped SHM their sum remains constant.

Final answer: K max/U min at equilibrium; U max/K zero at extremes; total energy constant without damping.

Question 19

Working: Total energy E=0.5kA^2=0.5(50)(0.2^2)=1 J. At x=0.1 m, U=0.5kx^2=0.5(50)(0.1^2)=0.25 J.

Final answer: E=1 J; U=.25 J.

Question 20

Working: Air resistance, friction and internal losses dissipate mechanical energy as heat and sound. The amplitude decreases (damping), so the oscillation eventually dies away.

Final answer: Damping transfers mechanical energy into heat/sound and reduces amplitude.